CSEC PHYSICS

JANUARY 2024 – Paper 2

Solutions

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SECTION A

Question 1.

Figure 1 shows a copper rod moving through a uniform magnetic field. The symbol (⊙) represents the magnetic field coming out of the page. Each end of the rod is connected to leads that are connected to an ammeter, which measures the electric current. The rod is only allowed to move in a horizontal direction (left to right/right to left) while in the magnetic field.

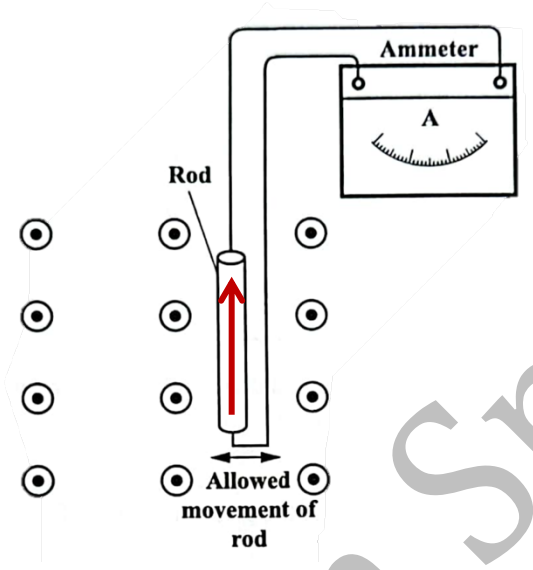

Figure 1. A copper rod moving through a uniform magnetic field

(a)

(i) Draw an arrow on the rod to indicate the direction of the induced current, when the rod moves to the left.

(1mark)

Answer is shown on diagram in red arrow drawn

(ii) Name the rule used to determine the direction of the induced current. **(1 mark)**

Fleming's Right-Hand rule

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(b) An experiment was performed to determine how the strength of the magnetic field affects the value of the induced current. Readings of the speed of movement of the rod, (u), and the induced current, (I), were obtained and recorded in Table 1.

TABLE 1: READINGS OF THE SPEED OF MOVEMENT OF THE ROD, (u) AND THE INDUCED CURRENT, (I)

Speed of Movement, u (ms^{-1})	Induced Current, I (A)	Induced emf, V (V)
0.1	0.0045	0.113
0.2	0.0110	0.275
0.4	0.0215	0.538
0.5	0.0270	0.675
0.6	0.0300	0.750
0.7	0.0380	0.950

(i) Given that the rod has a resistance of 25 Ω, use Ohm's law to complete Table 1 by calculating the induced emf, V. (5 marks)

Ohms Law: $V = IR$

When $I = 0.0045\ A$

$V = (0.0045) \times (25)$
$V = 0.1125$
$V = 0.113$ to 3 s.f.

When $I = 0.0215\ A$

$V = (0.0215) \times (25)$
$V = 0.5375$
$V = 0.538$ to 3 s.f.

When $I = 0.0300\ A$

$V = (0.0300) \times (25)$
$V = 0.750$

(ii) Use the results from Table 1 to plot a graph of induced emf, V, (V) against speed of movement, u, (ms^{-1}) on the grid provided in Figure 2 on page 7. Draw the line of best fit through the points. (8 marks)

Points to plot on the graph:

(0.1, 0.113), (0.2, 0.275), (0.4, 0.538), (0.5, 0.675), (0.6, 0.750), (0.7, 0.950)

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Graph of induced emf, V (V) against Speed of movement, u (ms^{-1})

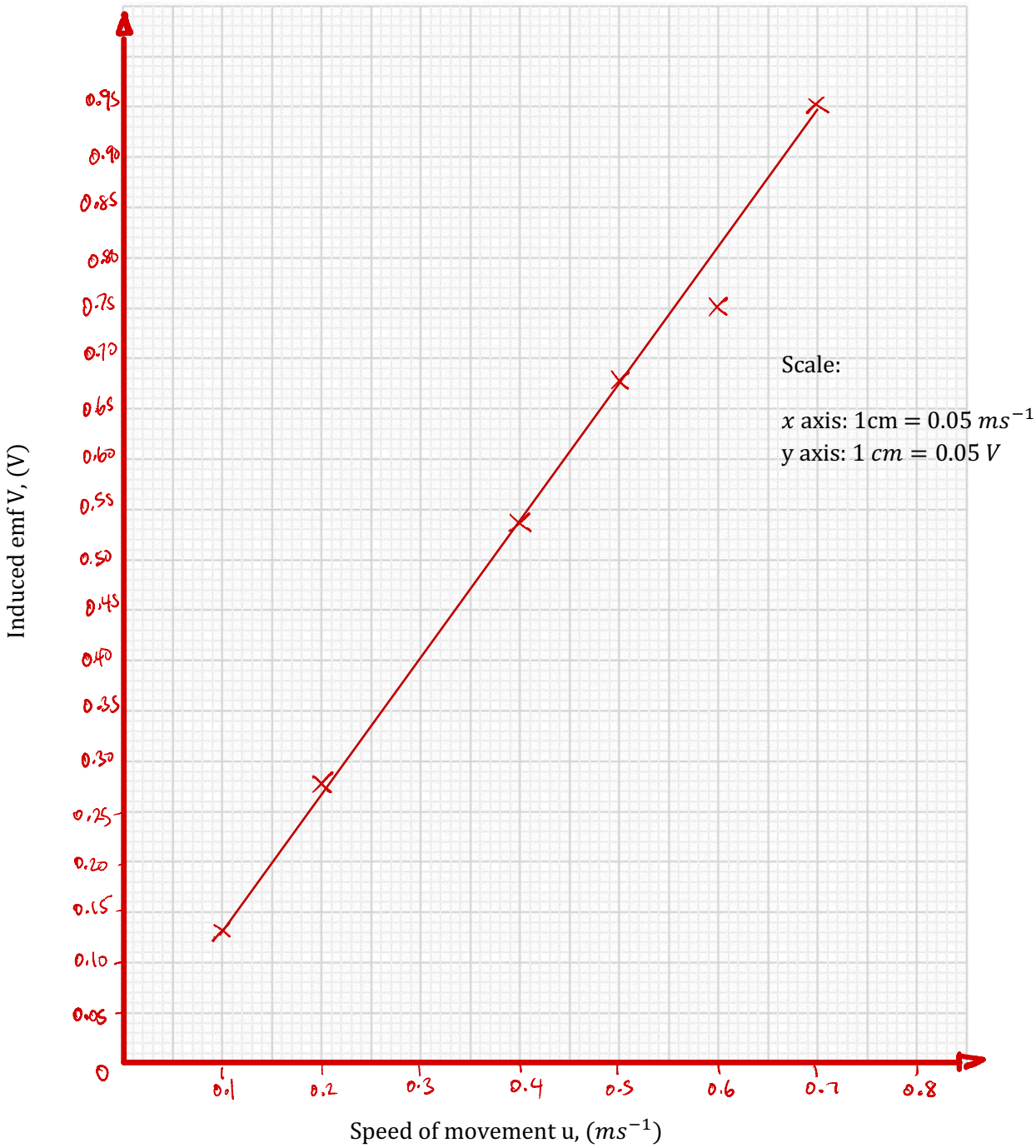

Figure 2. Graph of induced emf, V (V) against Speed of movement, u (ms^{-1})

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(iii) Use the graph to determine the gradient, G.

Using points on line of best fit: (0.15, 0.20) and (0.65, 0.88)

Gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{0.88 - 0.20}{0.65 - 0.15}$

$= 1.36\ Vsm^{-1}$

(5 marks)

(iv) The gradient, G, is related to the length of the rod, L, by the equation $G = 4.3L$. Calculate the length of the rod, L, using the answer obtained in (b) (iii).

$G = 4.3L$

So $L = \frac{G}{4.3}$

$L = \frac{1.36}{4.3}$

$L = 0.316\ m$

Therefore, the length of the rod is 0.316 m

(4 marks)

(c) The experiment shows that increasing the speed of movement can increase the induced emf. Give ONE other factor which would increase the induced emf in the rod.

Increase the strength of the magnetic field. (More magnets, Stronger magnets)

OR

Increase the length of the rod

Total 25 marks

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Question 2.

(a) (i) State the principle of conservation of energy. **(2 marks)**

Energy cannot be created or destroyed but can only be transformed from one form to another. The total energy of a closed system remains constant.

(ii) Define the term 'momentum'. State its SI units. **(2 marks)**

Momentum is the product of the mass and velocity of a body.

SI unit: kg m s^{-1}

b) State the energy changes that take place from the time the girl begins to exert a force on the shot put, until the shot-put hits the ground and stops moving. **(3 marks)**

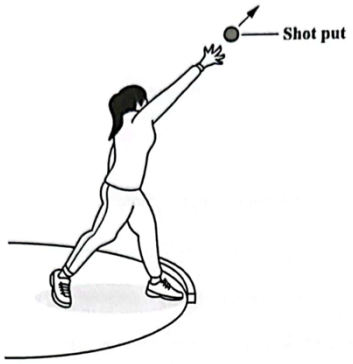

Figure 3. A girl throwing a shot put

Before the throw the shot put contains **gravitational energy** on account of being held a height off the ground.

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Chemical energy in the muscles of the girl is converted into **kinetic energy** of the shot-put. As the shot-put rises, some of the kinetic energy is converted into **gravitational potential energy**. When the shot-put falls, **gravitational potential energy** is converted back into **kinetic energy**.

When it hits the ground and stops, the kinetic energy is converted into **heat/thermal energy** and **sound**.

(c) The mass of the shot put is 4.0 kg. The girl exerts a force on the shot put for 0.60 s. The speed of the shot put increases from 0 m s^{-1} to 9 m s^{-1} before it leaves the girl's hand.

Calculate the

(i) Momentum of the shot put on leaving the girl's hand **(2 marks)**

$$Mass, m = 4.0\ kg\ Velocity, v = 9\ m/s$$

$$p = mv$$

$$p = 4.0 \times 9$$

$$p = 36\ kg\ m\ s^{-1}$$

(ii) Acceleration of the shot put **(3 marks)**

$$Initial\ velocity, u = 0\ ms^{-1} \qquad Final\ Velocity, v = 9\ ms^{-1}$$

$$a = \frac{(v - u)}{t}$$

$$a = \frac{(9 - 0)}{0.6}$$

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$$a = 15\ ms^{-2}$$

(iii) Average force exerted **(3 marks)**

$$F = ma$$

$$F = 4.0 \times 15$$

$$F = 60\ N$$

Total 15 marks

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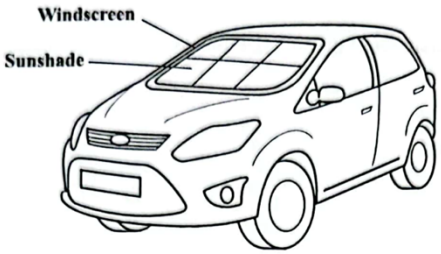
Question 3.

Figure 4 shows a car on a sunny day in a Caribbean country. A tourist placed a sunshade inside the windscreen to reduce the temperature rise of the air inside the car.



(a) Which material, dull black, dull white, shiny black, shiny white, is MOST suitable for the outer surface of the sunshade. Justify your choice. **(2 marks)**

Shiny white.
Shiny white surfaces are good reflectors and poor absorbers of heat radiation, so it reflects most of the solar radiation and reduces heating inside the car.

(b) Complete the scenarios by inserting the word which BEST describes the main process of thermal energy transfer. **(3 marks)**

i) A girl goes for a walk on a cold day. She touches an iron gate that removes thermal energy from her hand. This energy is then transferred through the iron gate by conduction.

ii) The girl holds the sides of a cup containing a hot drink. Her hands gain thermal energy by conduction.

iii) Some farm workers light a fire. The girl warms her hands by the side of the fire. Her hands gain thermal energy by radiation.

Should be noted that since the girl's hands was at the side and not over the fire the dominant form of energy transfer here is radiation and not convection.

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c) Define the term 'specific latent heat of fusion of a substance'. **(3 marks)**

The specific latent heat of fusion of a substance is the amount of heat energy required to change 1 kg of the substance from solid to liquid at its melting point without any change in temperature.

(d) Small pieces of ice with a temperature of 0 °C are added to 0.35 kg of water. The initial temperature of the water is 24.5 °C. The water loses 35 000 J of thermal energy as the temperature of the water decreases to 0 °C.

Calculate the

(i) specific heat capacity of water **(4 marks)**

$Mass\ of\ water\ =\ 0.35\ kg$

$Change\ in\ temperature\ =\ 24.5\ -\ 0$

$=\ 24.5\ K$

$Energy\ lost\ =\ 35\ 000\ J$

$$E\ =\ mc\Delta T$$

$$c = \frac{E}{m\Delta T}$$

$$c\ =\ \frac{35000}{(0.35 \times 24.5)}$$

$$c\ =\ \frac{35000}{8.575}$$

$$c\ \approx\ 4080\ J\ kg^{-1}\ °C^{-1}$$

(ii) mass of ice added to the water if all the ice melts. **(3 marks)**

Energy lost by water = energy gained by ice

$$Q\ =\ mL$$

$$m = \frac{Q}{L}$$

$$m\ =\ \frac{35000}{(3.3 \times 10^5)}$$

$$m\ \approx\ 0.106\ kg$$

Total 15 marks

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SECTION B

Answer ALL questions.

Question 4.

(a) Visible light, radio waves, X-rays, infrared, γ-radiation and ultraviolet radiation are types of electromagnetic waves.
Name the electromagnetic waves in EACH of the regions labelled A, B, C, D in the electromagnetic spectrum shown in Figure 5.

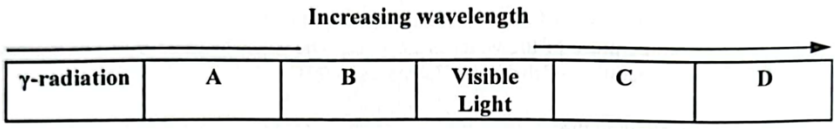

Figure 5. Electromagnetic spectrum

A X-rays

B Ultraviolet

C Infrared

D Radio waves **(4 marks)**

(b) State which sound wave parameters affects EACH of the following properties of a musical note.

(i) Pitch

Frequency

(ii) Loudness

Amplitude **(2 marks)**

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(c) A person standing 100 m from a wall claps his hands and hears an echo 0.6 s later. Calculate the speed of sound in air. **(4 marks)**

$Distance\ travelled\ by\ the\ sound\ =\ 2\ \times\ 100\ m\ =\ 200\ m$

$Time\ taken\ =\ 0.6\ s$

$$Speed\ =\ \frac{distance}{time}$$

$$Speed\ =\ \frac{200}{0.6}$$

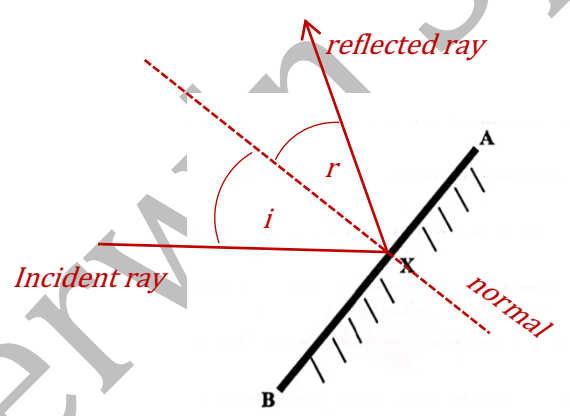
$$Speed\ =\ 333.3\ m\ s^{-1}$$

Therefore, the speed of sound in air is 333 m s^{-1}.

(d) Figure 6 shows a plane mirror AB. A ray of light strikes the mirror at point X with an angle of incidence, i, between 0° and 90°.

Complete the diagram to show the path of the incident ray and the corresponding reflected ray.

Mark on the diagram the angle of incidence, i, and the angle of reflection, r. **(5 marks)**



Note: Angle i = angle r

Total 15 marks

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Question 5.

(a) Differentiate between alternating current (ac) and direct current (dc). **(2 marks)**

Alternating current (ac) changes direction and magnitude periodically, whereas direct current (dc) flows in one direction only and is usually constant.

(b) (i) Draw the electrical symbol for a transformer. **(1 mark)**

(ii) State TWO advantages of using alternating current (ac) instead of direct current (dc) for transmitting electrical energy.
(2 marks)

1. AC can be easily stepped up or down using transformers for different voltage applications.

2. High voltage transmission reduces energy losses in power lines.

(iii) Complete the statement below.

Transformers work on the principle of mutual Electromagnetic Induction **(1 mark)**

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(c) A mobile phone charger contains a step-down transformer which converts the input of 220 V to a working output voltage of 5 V.

(i) If the output current that flows into the mobile phone is 120 mA, calculate the input current from the ac supply. **(3 marks)**

$I_s = 120mA$

$I_s = 0.120A$

$V_s = 5\ V$

$V_p = 220\ V$

For an ideal transformer:

$Power\ in\ primary\ coil\ =\ Power\ in\ secondary\ coil$

$VpIp\ =\ VsIs$

$I_p = \frac{V_s I_s}{V_p}$

$I_p\ =\ \frac{5 \times 0.120}{220}$

$I_p\ =\ 0.60\ /\ 220$

$I_p\ =\ 0.00273\ A$

OR $I_p\ =\ 2.73\ mA$

(ii) Determine the power supplied by the secondary turns. State your answer in SI units. **(4 marks)**

P = VI

P = 5 × 0.120

P = 0.60 W

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(iii) If there was a 10% loss of power within the transformer, calculate the actual power delivered to the secondary circuit? **(2 marks)**

Power supplied by the secondary = 0.60 W

10% loss = $0.10 \times 0.60 = 0.06$ W

Actual power delivered = 0.60 - 0.06

Actual power delivered = 0.54 W

Total 15 marks

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Question 6.

(a) The cloud chamber is a device used to detect the three types of radioactive emissions, alpha, beta and gamma. Figure 7 shows a cross section of three identical cloud chambers containing the different sources.

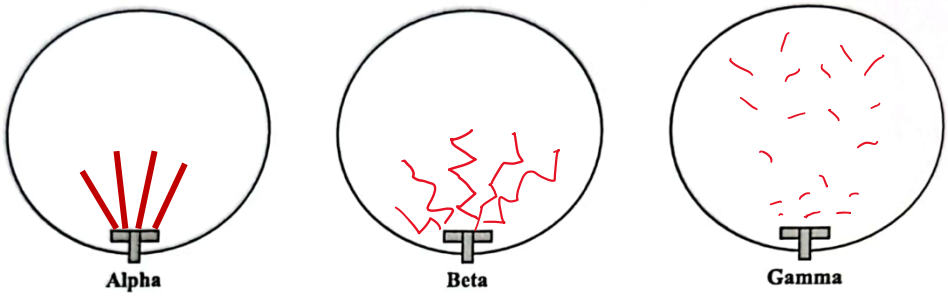

Figure 7. Cross section of three identical cloud chambers containing alpha, beta and gamma sources

(i) Illustrate the appearance of the tracks of radioactive emission in the cloud chambers in Figure 7.

(3 marks)

(ii) State THREE properties of gamma emissions. **(3 marks)**

1. Gamma rays are electromagnetic waves.
2. Can travel in a vacuum (and other properties of electromagnetic radiation)
3. Has no charge
4. High penetrating power as compared to other emissions

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b. Figure 8 shows the graph of the mass of a radioactive isotope against time elapsed.

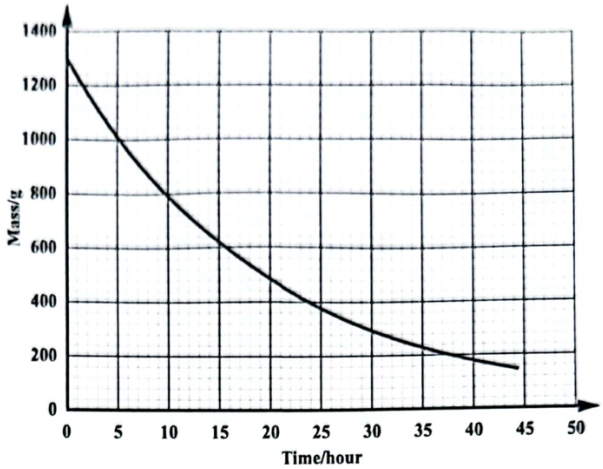

Figure 8. Graph of the mass of a radioactive isotope against time elapsed.

Use the graph in Figure 8 to

i. Determine the initial mass of the radioactive isotope
(1 mark)

From graph, Initial mass (when $t = 0$) is 1300 g

ii. Calculate the average half life of the radioactive isotope. Show all working.
(5 marks)

Half-life = time taken for mass to reduce to half its value

$Calculation\ 1$:

When mass $= 1300\ g$ time $= 5\ hours$

At Half mass $= 650\ g$, time $= 18\ hours$

$\therefore$ Half life $= 18 - 5 = 13\ hours$

$Calculation\ 2$:

When mass $= 400\ g$ time $= 24\ hours$

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Half mass $= 200\ g, time = 38\ hours$

$\therefore$ Half life $= 38 - 24 = 14\ hours$

$$Average\ half\ life = \frac{13 + 14}{2}$$

$$= 13.5\ hours$$

(c) The half-life of another radioactive nuclide is 10 hours. Determine the fraction of this radioactive nuclide that remains after 30 hours. **(3 marks)**

Answer:

Half-life = 10 hours

Time elapsed = 30 hours

Number of half-lives $= \frac{30}{10} = 3$

After each half-life, the quantity is halved:

After 1 half-life $\rightarrow \frac{1}{2}$

After 2 half-lives $\rightarrow \frac{1}{4}$

After 3 half-lives $\rightarrow \frac{1}{8}$

Therefore, the fraction remaining $= \mathbf{\frac{1}{8}}$

Total 15 marks

END OF TEST
IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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